

Complete classification of adjointable closed-submodule preservers

Candidate full-result packet for `fa_banach_001`
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Abstract

For an arbitrary right Hilbert $\mathcal{A}$ -module E , put $I_E = \overline{\text{span}}\langle E, E \rangle$. We prove that an adjointable operator T leaves every norm-closed right $\mathcal{A}$ -submodule invariant if and only if

$$Tx = xz \quad (x \in E)$$

for a unique $z \in Z(M(I_E))$. No injectivity, surjectivity, nonzero, unitality, finite generation, or orthonormal-vector hypothesis is needed. For full modules this is exactly multiplication by $Z(M(\mathcal{A}))$. Consequently, after the necessary nonzero repair, Sharifi's requested coefficient belongs to $\mathcal{A}$ for every full module precisely when $\mathcal{A}$ is unital. The proof turns closed submodules into closed right ideals of $\mathcal{K}(E)$, proves that a multiplier preserving every closed right ideal is central, and transfers the center through the Hilbert-module Morita equivalence.

1 Source question and normalization

Sharifi's Problem 2.16 in arXiv:2506.01161, PDF page 11, asks for which C^* -algebras universal closed-submodule invariance forces a bounded adjointable operator to be a coefficient morphism $I.a$ with nonzero $a \in \mathcal{A}$ [1].

INVARIANT SUBMODULES OF MODULAR OPERATORS

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We know that if every subspace of a Hilbert space is invariant under a linear transformation T , then T is a scalar transformation. This fact motivates us to close this section with the following problem. We will study such C^* -algebras and Hilbert C^* -modules elsewhere.

Problem 2.16. *Let E be a Hilbert $\mathcal{A}$ - $\mathcal{A}$ -bimodule. For which C^* -algebras $\mathcal{A}$ does the following assertion hold?*

If every closed submodule of E is invariant under a bounded adjointable operator $T : E \rightarrow E$, then T is a C^ -coefficient morphism, i.e., there exists a nonzero element $a \in \mathcal{A}$ such that $T = I.a$, where I is the identity operator on E .*

3. C^* -ALGEBRAS OF COMPACT OPERATORS

Suppose that $\mathcal{A}$ is an arbitrary C^* -algebra of compact operators, i.e. $\mathcal{A}$ is a C^* -subalgebra

The displayed source statement has two necessary normalizations. First, the zero operator preserves every submodule, so either $T \neq 0$ (as in the intended injective reading used by Frank) must be assumed or the zero coefficient must be allowed. Second, for nonunital $\mathcal{A}$ the identity operator is implemented by

$1_{M(\mathcal{A})}$, which need not belong to $\mathcal{A}$. The natural coefficient space is therefore the center of the multiplier algebra. The theorem below gives the exact answer, including non-full modules.

Frank's later arXiv:2507.11206 explicitly cites Problem 2.16. Its Theorem 1.2 proves the conclusion for *bijective bounded module morphisms* on full modules, and the paper states that the merely injective bounded-module case is out of reach [2]. The result below does not conflict with that boundary: it uses the source problem's stronger *adjointability* hypothesis in an essential way.

2 Idea of the proof

The compact-operator algebra $B = \mathcal{K}_{I_E}(E)$ is the correct surrogate for the scalar field. Every closed submodule $M \subset E$ determines the closed right ideal of compact operators whose ranges lie in M , and every closed right ideal arises this way. Thus the hypothesis says that the multiplier of B induced by T preserves every closed right ideal of B .

The key rigidity step is purely C^* -algebraic. If a multiplier b of a C^* -algebra preserves every closed right ideal, then every open support projection in the bidual is invariant under b . Were an irreducible image of b nonscalar, Kadison transitivity would produce a positive element whose support contains a vector ξ but not $b\xi$, contradicting that invariance. Hence b is central. Finally, the centers of the two multiplier algebras on opposite sides of a full Hilbert module agree: centrality on $\mathcal{K}(E)$ is equivalent to right multiplication by a unique element of $Z(M(I_E))$. We include direct proofs of every module-theoretic passage.

3 Three structural lemmas

Throughout, Hilbert-module inner products are linear in the second variable, and $\theta_{x,y}(w) = x\langle y, w \rangle$.

Lemma 1 (Submodules and right ideals). *Let E be a full right Hilbert I -module and put $B = \mathcal{K}_I(E)$. For a closed right ideal $J \subset B$, set*

$$M_J = \overline{\text{span}}\{Sx : S \in J, x \in E\}.$$

Then

$$J = \mathcal{K}(E, M_J) := \overline{\text{span}}\{\theta_{m,y} : m \in M_J, y \in E\}.$$

In particular, if $U \in \mathcal{L}_I(E)$ leaves every closed submodule invariant, then the multiplier of B induced by U leaves every closed right ideal of B invariant. The underlying submodule/right- M -ideal correspondence is also recorded in [6, Theorem 6.6].

Proof. The space M_J is a closed I -submodule. If m is a limit of finite sums of vectors Sx with $S \in J$, then $\theta_{Sx,y} = S\theta_{x,y} \in J$; hence $\mathcal{K}(E, M_J) \subset J$. Conversely, take a finite-rank approximate identity (e_λ) for B . If $S \in J$, then $Se_\lambda \rightarrow S$, and every Se_λ is a finite sum of operators $\theta_{Sx,y}$ with $Sx \in M_J$. Thus $J \subset \mathcal{K}(E, M_J)$.

Now suppose $U(M_J) \subset M_J$. For $S \in J$, the operator US is compact and has range in M_J . Applying the same finite-rank approximation to US shows $US \in \mathcal{K}(E, M_J) = J$. Under the standard identification $\mathcal{L}_I(E) = M(B)$, this says precisely that the corresponding multiplier left-multiplies J into itself. $\square$

Lemma 2 (Right-ideal rigidity). *Let B be a C^* -algebra and $b \in M(B)$. If*

$$bJ \subset J$$

for every norm-closed right ideal J of B , then $b \in Z(M(B))$.

Proof. Let $a \in B_+$, let $p = s(a) \in B^{**}$ be its support projection, and let $J = \overline{aB} = pB^{**} \cap B$. The weak-star closure of J is pB^{**} . These standard support-projection facts are recalled, for example, in [4]. The hypothesis and weak-star continuity of left multiplication therefore give

$$b(pB^{**}) \subset pB^{**}, \quad (1-p)bp = 0. \tag{1}$$

Let $\pi : B \rightarrow \mathcal{B}(H)$ be an irreducible nondegenerate representation, extended to $M(B)$ and normally to B^{**} . Suppose $\pi(b)$ is not scalar. Then some unit vector ξ has

$$\pi(b)\xi = \lambda\xi + \mu\zeta, \quad \zeta \perp \xi, \quad \mu \neq 0.$$

Kadison transitivity [5] supplies a self-adjoint $c \in B$ whose restriction to $\text{span}\{\xi, \zeta\}$ is the orthogonal projection onto $\mathbb{C}\xi$. Put $a = c^2$ and $p = s(a)$. Then

$$\pi^{**}(p)\xi = \xi, \quad \pi^{**}(p)\zeta = 0.$$

Applying π^{**} to (1) forces $\pi(b)\xi$ to lie in $\pi^{**}(p)H$, contradicting its nonzero ζ component. Hence $\pi(b)$ is scalar for every irreducible π .

For $x \in B$, the commutator $[b, x]$ belongs to B and is annihilated by every irreducible representation. Irreducible representations separate the points of a C^* -algebra, so $[b, x] = 0$ for all $x \in B$. Thus $b \in Z(M(B))$. $\square$

Lemma 3 (Transfer of the center). *Let E be a full right Hilbert I -module. If $U \in \mathcal{L}_I(E)$ commutes with $\mathcal{K}_I(E)$, then there is a unique $z \in Z(M(I))$ such that $Ux = xz$ for every $x \in E$.*

Proof. Write $I_0 = \text{span}\langle E, E \rangle$. For $a = \sum_k \langle x_k, y_k \rangle \in I_0$, define

$$L_U(a) = \sum_k \langle x_k, U y_k \rangle. \quad (2)$$

This is well-defined. Indeed, for $w \in E$,

$$wL_U(a) = \sum_k \theta_{w, x_k} U y_k = \sum_k U \theta_{w, x_k} y_k = U(wa). \quad (3)$$

If $a = 0$, the right side vanishes for every w ; fullness makes the right action of I faithful, so $L_U(a) = 0$.

Fullness also gives $\|d\| = \sup_{\|w\| \leq 1} \|wd\|$ for $d \in I$. Consequently (3) yields

$$\|L_U(a)\| \leq \|U\| \|a\|.$$

Thus L_U extends boundedly to I , and (2) shows $L_U(ab) = L_U(a)b$. Perform the same construction for U^* and set

$$R_U(a) = L_{U^*}(a^*)^*.$$

For $a = \langle x, y \rangle \in I_0$ and $b \in I$, equation (3) gives

$$aL_U(b) = \langle x, U(yb) \rangle = \langle U^*x, y \rangle b = R_U(a)b.$$

Density extends this identity to all $a, b \in I$. Hence (L_U, R_U) is a double centralizer, represented by a unique $z \in M(I)$ with $L_U(a) = za$ and $R_U(a) = az$.

Let (u_λ) be an approximate identity of I . Equation (3), first on I_0 and then by continuity, gives

$$wzu_\lambda = wL_U(u_\lambda) = U(wu_\lambda).$$

Passing to the limit yields $Uw = wz$. Applying this identity to wa and using I -linearity of U gives

$$w(az - za) = 0 \quad (w \in E, a \in I).$$

Faithfulness implies $az = za$ for every $a \in I$, so $z \in Z(M(I))$. If $Ez = 0$, then $Iz = 0$ by fullness and hence $z = 0$, proving uniqueness. $\square$

4 Exact classification

Theorem 1. *Let $\mathcal{A}$ be an arbitrary C^* -algebra, let $E \neq 0$ be a right Hilbert $\mathcal{A}$ -module, and put*

$$I_E = \overline{\text{span}}\langle E, E \rangle \triangleleft \mathcal{A}.$$

For $T \in \mathcal{L}_{\mathcal{A}}(E)$, the following are equivalent:

1. *every norm-closed right $\mathcal{A}$ -submodule of E is T -invariant;*
2. *there is a unique $z \in Z(M(I_E))$ such that $Tx = xz$ for every $x \in E$.*

Here the nondegenerate right I_E -action on E is extended canonically to $M(I_E)$.

Proof. The module E is full over I_E . Moreover, its closed $\mathcal{A}$ -submodules and closed I_E -submodules coincide. Only one direction needs comment: if M is a closed I_E -submodule, (u_λ) is an approximate identity of I_E , $x \in M$, and $a \in \mathcal{A}$, then

$$xa = \lim_{\lambda} (xu_\lambda)a = \lim_{\lambda} x(u_\lambda a) \in M,$$

because $u_\lambda a \in I_E$. Thus M is an $\mathcal{A}$ -submodule.

Assume (1) and put $B = \mathcal{K}_{I_E}(E)$. Adjointability identifies T with a multiplier $b \in M(B)$: left multiplication on B is $S \mapsto TS$, and right multiplication is $S \mapsto ST$ [3]. By Lemma 1, b preserves every closed right ideal of B . Lemma 2 gives $b \in Z(M(B))$, so T commutes with B . Lemma 3 now supplies a unique $z \in Z(M(I_E))$ such that $Tx = xz$.

Conversely, let $z \in Z(M(I_E))$. The action of every $a \in \mathcal{A}$ on the ideal I_E is a multiplier, so it commutes with z . Hence $R_z(x) = xz$ is $\mathcal{A}$ -linear and adjointable, with adjoint R_{z^*} . If M is a closed $\mathcal{A}$ -submodule and (u_λ) is an approximate identity of I_E , then

$$xz = \lim_{\lambda} x(zu_\lambda) \in M \quad (x \in M),$$

because $zu_\lambda \in I_E \subset \mathcal{A}$. Thus every closed submodule is R_z -invariant. □

Corollary 1 (Full modules and Sharifi's coefficient). *If E is full over $\mathcal{A}$, then the preservers in Theorem 1 are exactly*

$$\{R_z : z \in Z(M(\mathcal{A}))\}.$$

No injectivity is required. If T is nonzero (in particular, injective), its coefficient z is nonzero. The demand that every such coefficient belong to $\mathcal{A}$ for every full E holds exactly for unital $\mathcal{A}$.

Proof. Fullness says $I_E = \mathcal{A}$. If $\mathcal{A}$ is unital, then $M(\mathcal{A}) = \mathcal{A}$. Conversely, if $\mathcal{A}$ is nonunital, the identity operator on any full module preserves every submodule and corresponds to $1_{M(\mathcal{A})} \notin \mathcal{A}$. If it were R_a for some $a \in \mathcal{A}$, fullness would give $ba = b$ on a dense span and hence on all $b \in \mathcal{A}$, making a a unit, a contradiction. □

For a fixed non-full module, Theorem 1 is also an exact answer to the coefficient question: all preservers are implemented by elements of $Z(M(I_E))$, and all are implemented by coefficients from $\mathcal{A}$ precisely when the canonical map $\mathcal{A} \rightarrow M(I_E)$ contains this center in its range. The literal nonzero-coefficient version of the source statement still needs $T \neq 0$: on a nonzero full module, $T = 0$ cannot be implemented by a nonzero coefficient.

5 Verification and novelty bounds

The proof is analytic. The accompanying exact finite-matrix checker verifies the centrality conclusion in $M_d(\mathbb{Q})$ for $1 \leq d \leq 7$ and rechecks the earlier graph-submodule models; it is a regression test, not part of the proof. The separate file `verification.md` gives an adversarial audit of the ideal correspondence, open-projection argument, Morita-center transfer, and source quantifiers.

The local registry, solution, attempt, and proof-gap indexes were searched for arXiv:2506.01161, arXiv:2507.11206, closed-submodule preservers, closed right ideals, central multipliers, and one-sided M -ideals. Bounded web/arXiv searches used the close phrases “every closed submodule”, “adjointable”, “central multiplier”, and “every closed right ideal”. They located Frank’s bijective theorem, the standard submodule/right-ideal correspondence, and the operator-space fact that closed Hilbert-module submodules are complete right M -ideals [6], but no theorem stating the adjointable classification proved here. This is bounded novelty evidence, not a publication-novelty guarantee.

Human review should concentrate on Lemma 2, especially the passage from right-ideal invariance to $(1 - p)bp = 0$, and on the double-centralizer construction in Lemma 3. Subject to that review, the verdict is `candidate_full_result_likely_valid`.

References

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